

Waste Material Classification using MobileNetV2

Technical Analysis Document

Introduction to Computer Vision - Final Project

Project type: Multi-class image classification

Framework: Python / PyTorch / Torchvision

Architecture: MobileNetV2 with transfer learning

Target classes: Cardboard, Glass, Metal, Paper, Plastic

This document presents the problem statement, methodology, implementation, experimental results, failure analysis, interpretability, limitations, ethical considerations, and reproducibility of the Computer Vision project.

1. Problem Statement and Project Motivation

Waste segregation is a practical environmental problem in which discarded objects must be assigned to the correct material category before recycling or disposal. Manual sorting can be slow and inconsistent, while visual appearance varies because of object shape, illumination, background, deformation and contamination. The objective of this project is to develop a reproducible Computer Vision system that automatically classifies photographs of waste into five common material categories.

The system receives an RGB image and predicts one of five labels: cardboard, glass, metal, paper or plastic. The project covers the complete workflow from dataset preparation to quantitative evaluation and visual interpretation.

1.1 Objectives

- Prepare a deterministic training, validation and test split.
- Use learned CNN features through a pretrained MobileNetV2 backbone.
- Train a custom five-class classification head using transfer learning.
- Evaluate the system with Accuracy, Precision, Recall, F1-score and a Confusion Matrix.
- Provide single-image inference with confidence scores and complete class probabilities.
- Use Grad-CAM to inspect the image regions influencing predictions.

1.2 Dataset

The project uses the Garbage Dataset - A Comprehensive Image Dataset for Garbage Classification and Recycling. The source dataset contains 13,348 images across 10 classes: Metal (930), Glass (1,736), Biological (699), Paper (1,336), Battery (756), Trash (453), Cardboard (1,411), Shoes (1,449), Clothes (1,892), and Plastic (1,597). The dataset is distributed under the MIT License.

Five material-oriented classes were selected: Cardboard, Glass, Metal, Paper and Plastic. The preprocessing script creates a 70% training, 15% validation and 15% test split using random seed 42. The resulting held-out test set contains 1,056 images.

2. Methodology

2.1 End-to-End Pipeline

Data acquisition and preprocessing: original images are read from the raw dataset, validated and organized into training, validation and test directories. Training images are resized to 224 x 224 pixels and augmented using random horizontal flip, random rotation and ColorJitter. Validation and test images are resized without random augmentation. All images are converted to tensors and normalized using ImageNet statistics.

Feature representation: the project uses learned features rather than handcrafted descriptors. MobileNetV2 pretrained on ImageNet is used as the feature extractor. The convolutional backbone is frozen in the implemented transfer-learning stage and the original classifier is replaced with a custom head containing Dropout with probability 0.3 followed by a Linear layer mapping 1,280 features to five output classes.

2.2 Training Configuration

Component	Configuration
Architecture	MobileNetV2
Pretraining	ImageNet
Output classes	5
Image size	224 x 224
Batch size	32
Optimizer	Adam
Learning rate	0.001
Loss	CrossEntropyLoss
Classifier	Dropout(0.3) + Linear(1280, 5)
Random seed	42

2.3 Post-processing

During inference, network logits are converted to probabilities using softmax. The class with the highest probability is selected as the prediction. The prediction module reports the predicted class, confidence score and full probability distribution. Grad-CAM is then available as an interpretability stage to visualize influential regions.

3. Implementation and Reproducibility

The implementation is modular and divided into five Python scripts. `prepare_data.py` creates the dataset split and supporting summaries; `train.py` defines transformations, `DataLoaders`, `MobileNetV2`, loss function and optimizer, executes training and validation, and saves the best checkpoint; `evaluate.py` evaluates the held-out test set; `predict.py` performs single-image inference; `gradcam.py` generates visual explanations.

3.1 Repository Organization

`data/raw/original/` stores the source images locally; `data/processed/` stores train/validation/test subsets; `models/` stores the trained checkpoint; `results/` stores metrics, plots and visual explanations; `src/` contains the five executable scripts.

3.2 Reproducible Environment

A clean Python virtual environment was recreated and all project dependencies were installed successfully from `requirements.txt`. The complete pipeline was then validated. Dataset preparation, model evaluation, single-image prediction and Grad-CAM all completed successfully.

3.3 Validated Execution

- `python src/prepare_data.py`
- `python src/train.py`
- `python src/evaluate.py`
- `python src/predict.py`
- `python src/gradcam.py`

3.4 Example Inference

For the test image `plastic_1.jpg`, the system predicted plastic with 67.43% confidence. The remaining probabilities were glass 26.14%, cardboard 2.27%, metal 2.14% and paper 2.02%. This demonstrates that the inference stage reports both the final label and useful uncertainty information.

4. Training Behaviour

The saved training history contains one recorded epoch. For that epoch, training accuracy is approximately 68.8% and validation accuracy is 83.41%. Training loss is approximately 0.92 and validation loss is 0.5838. Because only one epoch is recorded, these plots should be interpreted as a snapshot rather than a convergence trend.

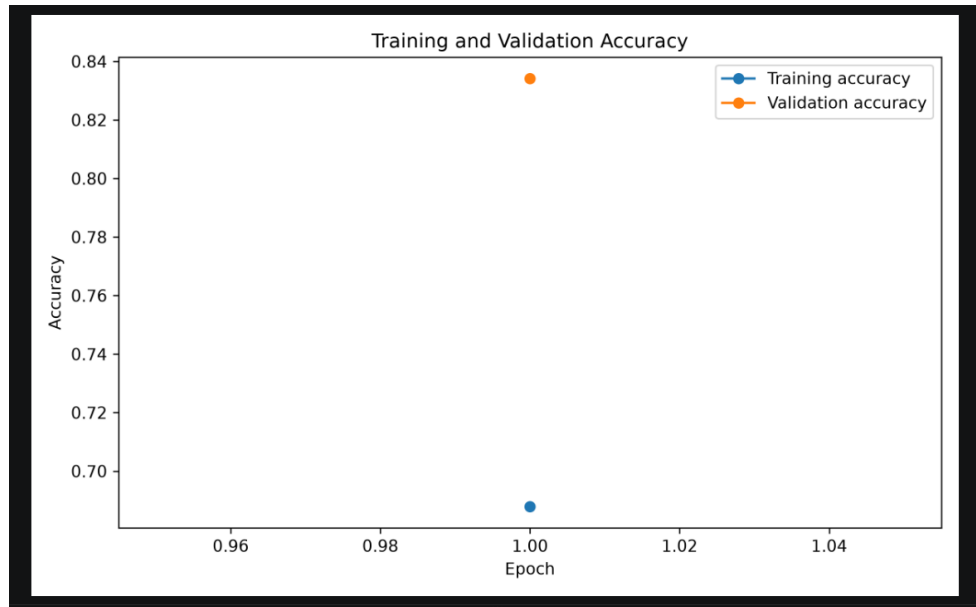


Figure 1. Training and validation accuracy from the recorded training history.

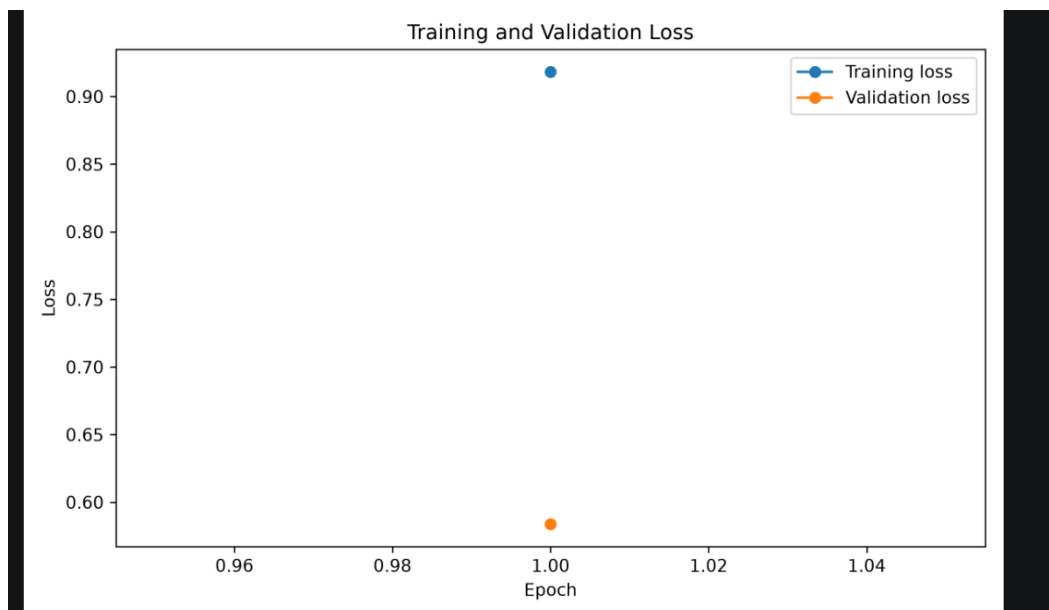


Figure 2. Training and validation loss from the recorded training history.

5. Experimental Results

The final trained model was evaluated on 1,056 unseen test images. The evaluation script computes Accuracy, Precision, Recall, F1-score, a complete classification report and a confusion matrix.

Metric	Result
Accuracy	82.39%
Macro Precision	81.94%
Macro Recall	82.68%
Macro F1-score	81.97%
Misclassified images	186 / 1056

An accuracy of 82.39% indicates that the model correctly classified more than four out of five unseen test images. The macro precision, recall and F1-score are close to the overall accuracy, suggesting reasonably balanced performance across the five selected categories.

5.1 Per-Class Performance

Class	Precision	Recall	F1-score	Support
Cardboard	0.8894	0.8310	0.8592	213
Glass	0.8450	0.8774	0.8609	261
Metal	0.6994	0.8643	0.7732	140
Paper	0.7877	0.8308	0.8087	201
Plastic	0.8756	0.7303	0.7964	241

Glass reaches the highest F1-score (0.8609), closely followed by cardboard (0.8592). Metal shows high recall (0.8643) but lower precision (0.6994), indicating that images from other classes are sometimes predicted as metal. Plastic has the lowest recall (0.7303), indicating a comparatively larger number of missed plastic samples.

5.2 Confusion Matrix Analysis

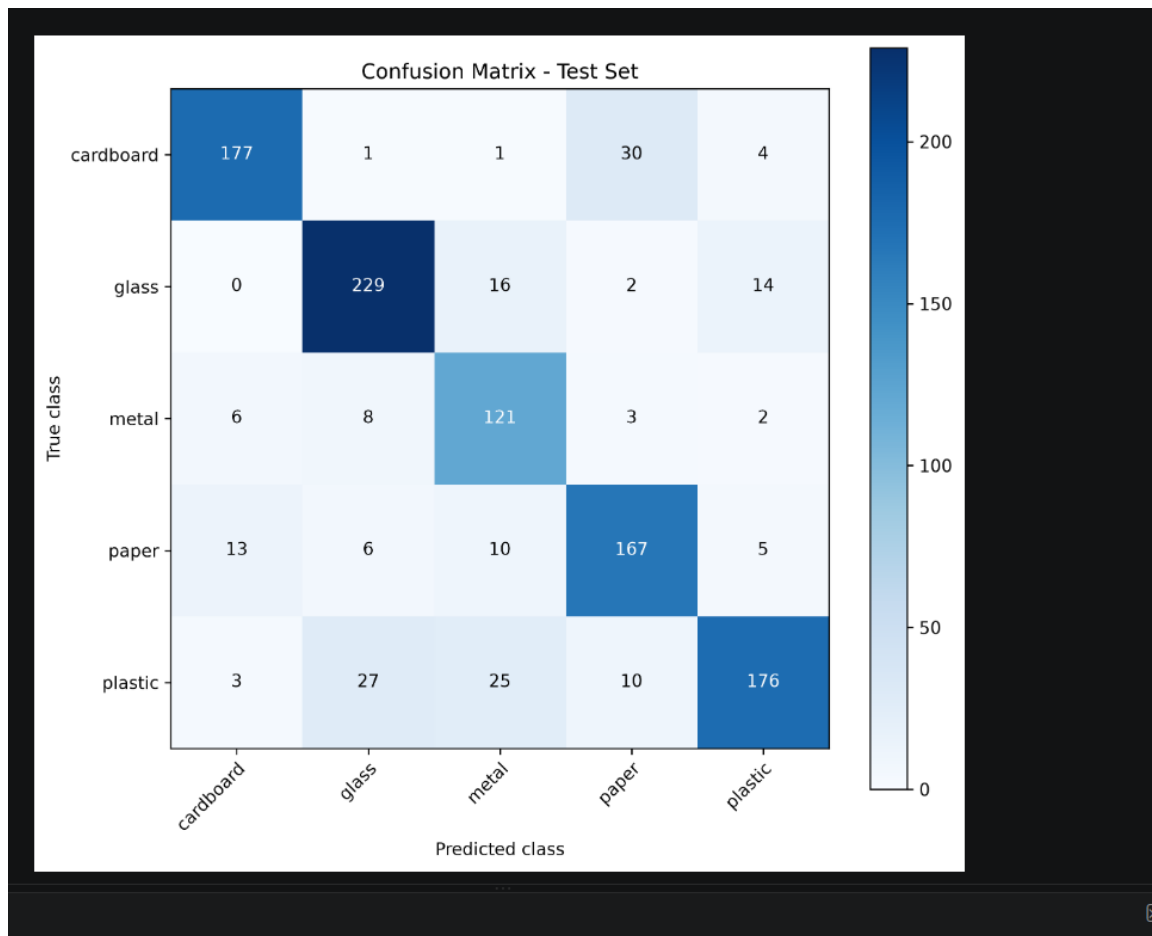


Figure 3. Confusion matrix on the 1,056-image held-out test set.

The largest single confusion is cardboard predicted as paper (30 samples). Plastic is frequently confused with glass (27 cases) and metal (25 cases). Glass is sometimes classified as metal (16) or plastic (14). These patterns are consistent with similarities in texture, transparency, reflections, shape and object appearance.

6. Failure Analysis

A total of 186 test images were misclassified. Cardboard and paper can share fibrous textures, flat geometry and printed surfaces. Transparent or reflective plastic may resemble glass, while reflective packaging can resemble metal. The classifier may also exploit contextual cues such as background and illumination rather than material composition alone.

The evaluation script saves `misclassified_images.csv` and `test_predictions.csv`, enabling systematic inspection of individual failure cases and supporting future error-driven improvements.

6.1 Main Limitations

- Only five of the ten available dataset categories are used.
- The classes are not perfectly balanced; metal has fewer source images than glass or plastic.
- The implemented transfer-learning configuration keeps the MobileNetV2 feature extractor frozen.
- Image-level classification assumes one dominant material per image.
- Performance on images captured outside the dataset distribution has not been measured.

6.2 Future Improvements

Future iterations could fine-tune later MobileNetV2 blocks, use class weighting, train on all ten categories, compare alternative architectures such as ResNet or EfficientNet, expand external testing, use real-time inference and develop deployment-oriented interfaces.

7. Interpretability with Grad-CAM

Grad-CAM provides a spatial explanation of the model prediction by highlighting regions that contribute most strongly to the selected class. The implementation targets the final convolutional block of MobileNetV2 and generates the original image, heatmap, overlay and combined visualization.

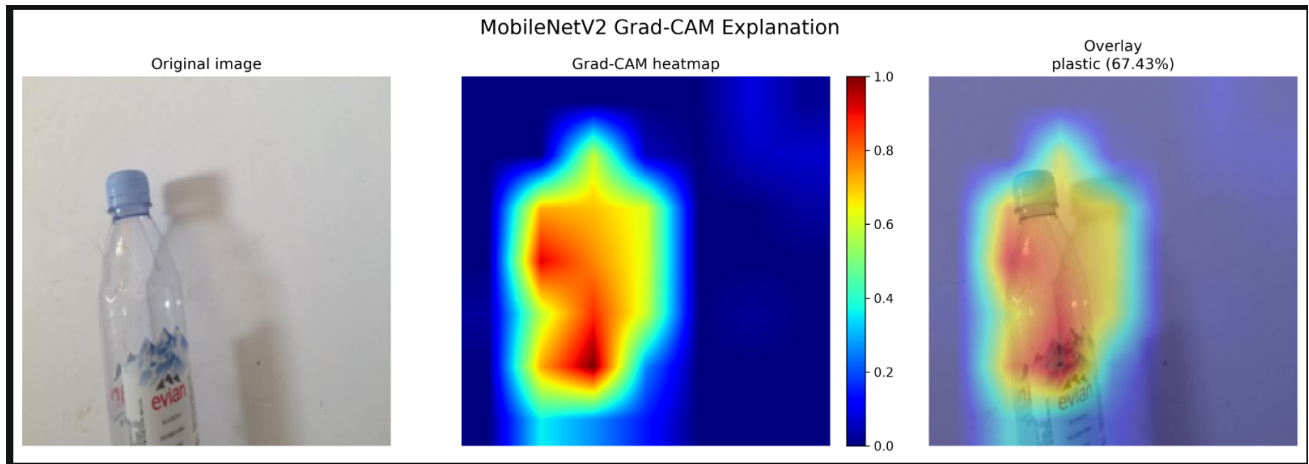


Figure 4. Real Grad-CAM output generated by the project for plastic_1.jpg. The model predicts plastic with 67.43% confidence. The heatmap and overlay show that the strongest activation is concentrated over the bottle body rather than being distributed uniformly across the background.

This example supports the interpretation that the model is responding to image regions associated with the object. Grad-CAM does not prove that the network has learned material composition itself, but it is useful for identifying whether predictions are driven by plausible object regions or by irrelevant background cues.

8. Ethical, Privacy and Security Considerations

The project processes waste-object images rather than identity, medical, financial or other directly sensitive personal information. Nevertheless, dataset provenance and licensing must be respected, and incidental background information in photographs should be considered when datasets are collected or redistributed.

Dataset bias remains important: lighting, camera type, geographical context, packaging design and waste condition may differ from the training distribution. Unequal class representation can also affect model behaviour. The reported 82.39% test accuracy is appropriate for an academic prototype but is not sufficient to justify autonomous industrial deployment without additional validation, monitoring, confidence thresholds and human review for uncertain predictions.

Transparency is improved by reporting class probabilities, confusion patterns and Grad-CAM visualizations. Reproducibility is supported by a fixed random seed, documented dependencies, modular scripts and saved evaluation outputs.

9. Conclusion

This project implements and validates a complete Computer Vision classification pipeline for five waste-material categories. It covers data preparation and augmentation, learned feature representation with MobileNetV2, a custom classification head, quantitative evaluation, inference, failure analysis and Grad-CAM interpretability.

The final model achieves 82.39% accuracy, 81.94% macro precision, 82.68% macro recall and 81.97% macro F1-score on 1,056 held-out test images. The analysis identifies meaningful confusion patterns between visually similar materials and provides clear directions for future improvement.

10. References

1. Garbage Dataset - A Comprehensive Image Dataset for Garbage Classification and Recycling. Dataset information supplied with the project source.
2. The Garbage Dataset (GD): A Multi-Class Image Benchmark for Automated Waste Segregation.
3. Managing Household Waste Through Transfer Learning.
4. Sandler, M. et al. MobileNetV2: Inverted Residuals and Linear Bottlenecks. CVPR, 2018.
5. Selvaraju, R. R. et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV, 2017.